retrieval. Although we've explored this scenario in the preceding chapter, the implementation part was very basic and manual. With LangChain, we can use the chaining functionality of the framework to create composable and reusable pipelines that create embeddings for our documents. The process of preparing documents for retrieval involves several key steps, as shown in figure 6.5.

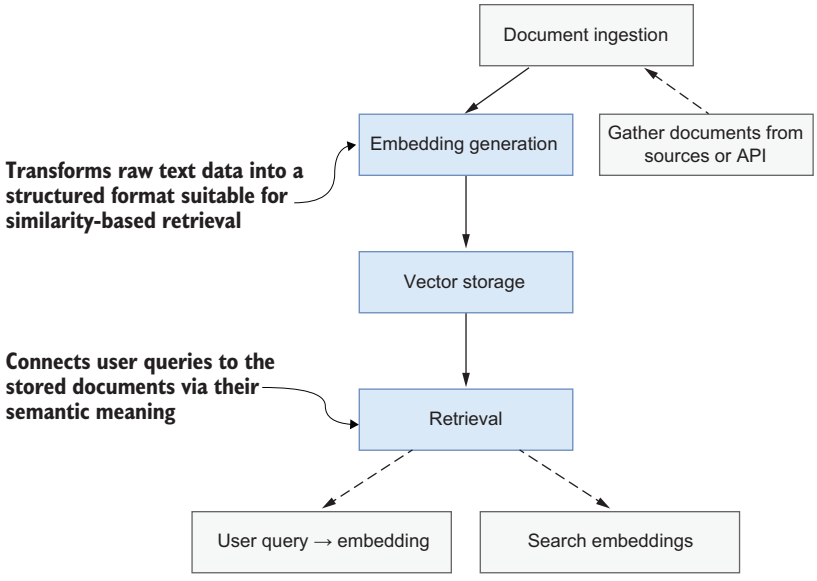



**Figure 6.5 The process of document embedding preparation and retrieval. It starts with document ingestion from various sources, followed by embedding generation using LangChain to create semantic vectors. These embeddings are stored in a vector database, enabling efficient similarity-based querying during the information retrieval stage.**

The main steps for preparing and retrieval of documents are

- *Document ingestion*—First, we need to gather our documents, which may come from various sources such as databases, APIs, or local files. We also expose an API endpoint to allow the user to upload their own documents from the UI.
- *Embedding generation*—Once the documents are ingested, we can use LangChain to generate embeddings for each document. These embeddings are vector representations that capture the semantic meaning of the content, allowing for effective similarity comparisons during retrieval.
- *Vector storage*—After generating embeddings, we need to store them in a suitable database or vector store. This storage solution should support efficient querying capabilities to allow quick access to the most relevant documents based on user input.